MIL-B-17228E
10 August 1984
SUPERSEDING
MIL-B-17228D
23 February 1972

MILITARY SPECIFICATION

BOILERS, STEAM AND HOT WATER; HIGH- AND LOW-PRESSURE, FORCED-CIRCULATION, PACKAGED TYPE (400,000 BTUS PER HOUR AND OVER OUTPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers compact, forced-circulation, coiled-tube type, packaged steam and hot water boilers with all accessories and controls necessary to form completely integrated, self-contained units ready for operation upon completion of service connections.

1.2 Classification.

1.2.1 Boiler types and classes. Boilers shall be of the following types and classes, as specified (see 6.2.1).

Type I - Light oil fired (grade 2).
Type II - Gas fired (natural, manufactured, mixed, or liquefied petroleum gas), as specified (see 6.2.1).
Type III - Combination light-oil and gas-fired (grade 2 oil, type of gas), as specified (see 6.2.1).

Class 1 - Low pressure (15 pound-force per square inch gage (psig), operating pressure).
Class 2 - High pressure (150, 200, 250, or 300 psig design pressure), as specified (see 6.2.1).
Class 3 - High-pressure hot water (160, 200, 250, 300, 400, or 500 psig design pressure), as specified (see 6.2.1).

1.2.2 Boiler sizes. For the purpose of defining requirements herein, boilers covered by this specification are divided into the following sizes based on the output in British thermal units per hour (Btu/hr) from and at

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4410-4520

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

212 degrees Fahrenheit (°F). The actual boiler capacity shall be as specified (see 3.8.1 and 6.2.1):

Size 1 - 400,000 to 2,500,000 Btu/hr.
Size 2 - 2,500,001 to 5,000,000 Btu/hr.
Size 3 - 5,000,001 to 8,000,000 Btu/hr.
Size 4 - 8,000,001 to 14,000,000 Btu/hr.
Size 5 - 14,000,001 Btu/hr and above.

1.2.3 Burner control systems. Burner control systems and combustion control systems (see 6.5.1) shall be of the following types as specified (see 6.2.1):

Programming control systems:

Control Sequence IA - Automatic, recycling direct electric ignition.
Control Sequence IB - Automatic, recycling with proved igniter.

Combustion control systems:

On-off.
High-Low-Off.
Modulating.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2.1), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-T-13867 - Treatment, Moisture- and Fungus-Resistant for Fire Control Electrical and Electronic Instruments and Equipment.
MIL-P-17749 - Pumping Units, Condensate, Heating and Pumping Units, Vacuum, Heating.
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility, Requirements for Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations (CFR)
Title 40, Part 60.344, Visible Emissions, Method 9.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Lexicon Steam Generating Equipment.
Measurement of Sound from Boiler Units, Bottom Supported, Shop or Field Erected.

(Application for copies should be addressed to the American Boiler Manufacturers Association, 950 North Glebe Road, Suite 160, Arlington, VA 22033.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip.
A450 - General Requirements for Carbon Ferritic Alloy and Austenitic Alloy Steel Tubes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

D396 - Fuel Oils.
D2156 - Test for Smoke Density in the Flue Gases from Distillate Fuels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code
Section I - Power Boilers.
Section IV - Heating Boilers.

Performance Test Code (PTC)
PTC 4.1-Steam-Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Factory Mutual Approval Guide.

(Application for copies should be addressed to the Factory Mutual Engineering Corporation, Factory Mutual System, 1151 Boston-Providence Turnpike, Norwood, MA 02062.)

INSTRUMENT SOCIETY OF AMERICA (ISA)

S18.1 - Standard for Annunciator Sequences and Specifications.

(Applications for copies should be addressed to the Instrument Society of America, P.O Box 12277, Research Triangle Park, NJ 27709.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
ICS 2 - Standards for Industrial Control Devices, Controllers, and Assemblies.
ICS 6 - Enclosures for Industrial Control Systems.
MG 1 - Standards for Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.
No. 85A - Prevention of Furnace Explosions in Fuel Oil and Natural Gas-fired Single Boiler Burner Furnaces.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 296 - Oil Burners.
UL 353 - Limit Controls.
UL 726 - Oil-fired Boiler Assemblies.
UL 795 - Commercial Gas Heating Equipment.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The steam-generating and hot water boilers shall consist of a forced-circulation, single- or multiple-coil water tube boiler with integrally mounted burner, draft fan, pumps, combustion and safety controls, control panel, and all appurtenances and boiler trim necessary to form a completely self-contained unit. The units shall be mounted on a base constructed of structural steel members, formed sheet steel, or a combination thereof.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a complete boiler for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standards compliance. Prior to approval of the first article, if one is specified, or prior to approval of the first shipment by the contractor if a first article is not specified, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the boiler(s) proposed to be furnished under this specification meet the applicable requirements of 3.4, 3.10.1, and 3.10.2 as applicable. Acceptable evidence of meeting the requirements of 3.4, 3.10.1 and 3.10.2 as applicable shall be either the UL label and ASME Stamp symbol of the applicable section of the ASME Boiler and Pressure Vessel Code, or the certification (see 6.2.2) by an authorized official of the contractor, stating that the boiler(s) have been tested and that the unit(s) meet the requirements of the specified standards in 3.4, 3.10.1, and 3.10.2 including specified methods of testings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.4 Design. The boilers shall be designed in accordance with the ASME Boiler and Pressure Vessel Code, section I, for class 2 and 3 high-pressure boilers and section IV for class 1, low-pressure boilers. The operating controls for the class 2 and 3 boilers shall be adjusted and set for the steam or water operating pressure in psig, steam temperature, in °F, feed water temperature °F, and temperature differential for hot water boiler between boiler discharge and system return as specified in the contract (see 6.2.1). The steam boilers shall be so designed that the rate of fluid flow through the steam-generating system is sufficient to insure that the temperature of the fluid in any part of the system does not exceed the saturation temperature for the specified working pressure during normal operation of the boilers. The hot water boilers shall be designed to meet the requirements of 3.8.5. The design of the combustion chamber and convection space shall facilitate free airflow without the possibility of gas pocketing. Boiler design shall permit accessibility for normal maintenance and service in the field.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7 Construction. The boiler shall be constructed for the specified design pressure in accordance with the applicable section and provisions of the ASME Boiler and Pressure Vessel Code and any current amendments thereto. The boilers shall carry the applicable ASME symbol stamps in accordance with 3.4.

3.7.1 Heat exchange coils. The heat exchange surface shall consist of a single, continuous coil or of multiple, header-connected coils arranged in a circular pattern. Provisions shall be made to allow for unequal expansion and contraction of the coil(s) under all firing conditions. The coil(s) shall be installed in a manner to permit ready removal and replacement. The water circulation system shall be equipped with a valve or valves to permit complete drainage by gravity or by blowing with compressed air of the coil(s) and steam separation vessel and with such additional connections as may be required to drain other piping, pumps, and water-containing accessories. The coils shall be fabricated of steel tubing approved under the applicable section of the ASME Boiler and Pressure Vessel Code. The tubing to be used for the coils

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

shall be mill-tested by the tubing manufacturer at a hydrostatic test pressure of at least 1000 psig in accordance with the applicable provisions of ASTM A450.

3.7.2 Combustion chamber. The combustion chamber shall be of such size as to provide ample volume for efficient combustion over the specified range of firing rates when the specified fuel is used. Heat distribution shall be uniform with no hot spots under all firing conditions. Combustion chamber walls shall be either water-cooled or air-cooled by directing the flow of combustion air between the inside of the boiler casing and the outside of the combustion chamber wall. The chamber shall be equipped with an observation port to permit inspection of the burner flame (and the pilot flame when used).

3.7.3 Pressure vessels. Boilers shall be furnished with a pressure vessel equipped for separating water and steam after the fluid leaves the heat exchange coils. Unless otherwise specified (see 6.2.1), the separator shall be designed to supply steam having a moisture content at all firing rates not exceeding one percent. Unless otherwise specified (see 6.2.1), the boiler water concentrations shall have not more than 3,500 parts per million of total solids. The steam separators shall be of the vertical type or horizontal drum type, at the option of the supplier.

3.7.3.1 Vertical separators. Vertical separators shall have sufficient capacity to handle a volume of water not less than 10 percent in excess of the volume evaporated at the maximum rated output of the boiler. A steam trap shall be installed to permit return of excess water back to the feedwater and condensate return system. Vertical separators shall be furnished with one blowdown valve, a manually-operated steam discharge valve, a steam safety valve, a steam pressure connection with gage, and additional appurtenances as required by the manufacturer's standard design to insure safe and efficient boiler operation.

3.7.3.2 Horizontal drum separators. Horizontal drum separators shall be mounted integrally with the unit and shall be equipped with a manually operated steam discharge valve, a steam safety valve, one blowdown valve, a steam lance, a feedwater connection, an automatic water level control capable of controlling operation of the feedwater pump, required steam pressure control switches, gage glass with try cocks, a steam pressure gage, and suitably located cleanout plugs. Unless otherwise specified (see 6.2.1), when a drum separator is furnished, a feedwater pump shall also be furnished. The feedwater pump shall have sufficient capacity to deliver twice the total quantity of water required by the boiler at maximum output and maximum operating pressure. The drum shall have sufficient water storage volume and steam separation space to permit response to load fluctuations within the limits specified herein.

3.7.4 Casing. The combustion and heat exchange chambers shall be fully jacketed by a steel casing. The casing shall be constructed to prevent combustion gas leakage and shall be designed to permit removal for maintenance, cleaning of heat exchange coils, and repair and replacement of any refractory. The casing shall be suitably insulated to prevent its outside temperature from exceeding 130°F at an ambient temperature of 80°F plus or minus 5°F. Insulation shall be held securely in place to withstand normal shocks and impacts associated with domestic and overseas transport.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.7.5 Steam and water connections. Unless otherwise specified (see 6.2.1), steam and water outlet connections shall be sized in accordance with the manufacturer's standard practice for the specified design pressure and shall either be flanged or threaded in accordance with the ASME Boiler and Pressure Vessel Code.

3.7.6 Boiler trim. Boiler fittings, pressure gages, feed stop and check valves, safety valves, and all other boiler trim and appurtenances shall comply with the applicable requirements and recommendations of the ASME Boiler and Pressure Vessel Code. The steam pressure gage shall be complete with siphon, gage cock, and inspector's test cock. A flue gas dial thermometer shall be suitably mounted in the flue gas outlet.

3.8 Performance. Unless otherwise specified (see 6.1.2 and 6.2.1), the performance requirements specified herein shall be based on the following criteria:

a. Fuel oil (percent by weight as fired):

Ultimate analysis:

Hydrogen	10.33
Carbon	87.87
Oxygen	0.25
Nitrogen	0.14
Sulphur	1.16
Ash	0.25

Higher heating value - - - - - - - 18,400 Btu per pound (Btu/lb).

b. Natural gas (percent by volume as fired):

Ultimate analysis:

Methane	90.0
Ethane	5.0
Nitrogen	5.0
Ash	Nil
Moisture	0.01

Higher heating value - - - - - - - 1,075 Btu per standard cubic foot (Btu/scf), 21,824 Btu/lb

The gas supply pressure at the site upstream from the service pressure regulator shall be as specified in the contract (see 6.2.1). When boilers are acquired for stock and the conditions at the site are not known in advance, a gas pressure at the inlet to the gas piping of 5 psig shall be assumed for design purposes.

3.8.1 Capacity. The boiler capacity in Btu/hr shall be as specified in the contract (see 1.2.2 and 6.2.1). Unless otherwise specified (see 6.2.1), the capacity shall be based on an operating pressure of 14.7 pound-force per square inch absolute (zero psig) and a feedwater temperature of 212°F (from and at 212°F). If the acquisition documentation should specify capacity in terms of boiler horsepower (hp) or pounds of steam per hour, the output shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

be computed on the basis of 33,475 Btu/hr per boiler hp or 970 Btu/hr per pound of steam from and at 212°F. The rated boilers capacity shall be attained under the specified conditions of steam outlet pressure and feedwater inlet temperature without exceeding the heat transfer rate specified in 3.8.3.

3.8.2 Efficiency.

3.8.2.1 Combustion efficiency. In accordance with ASME PTC 4.1, as determined by flue gas analysis, the levels for carbon dioxide, carbon monoxide, and oxygen shall be as specified in table I and shall be not less than the average of the three rated loads listed in table I. When fired with fuels specified herein the boiler shall not exceed the smoke density readings listed in table I when measured in accordance with ASTM D2156 (Shell-Bacharach) of the values except for short periods not to exceed 3 minutes. Both stable firing and combustion efficiency shall be maintained over the entire firing range required by table I. For On-Off and High-Low-Off controlled units, combustion efficiency shall not be less than the average of three maximum and minimum firings at rated loads.

3.8.2.2 Fuel-to-steam efficiency. In accordance with ASME PTC 4.1, using the input/output method, the fuel-to-steam efficiency for light oil fired boilers shall be a minimum of 80 percent and for gas fired boilers shall be a minimum of 78 percent when fired at maximum rated output established in accordance with the provisions of 3.8 and 3.8.1.

3.8.2.3 Fuel-to-hot water efficiency. In accordance with ASME PTC 4.1 using the input/output method, the fuel-to-hot water efficiency for class 3 boilers shall be a minimum of 80 percent when fired at maximum rated output established in accordance with the provisions of 3.8 and 3.8.1.

3.8.3 Heat transfer rate. When the boiler is operating at maximum rated capacity, the heat transfer rate through the boiler heating surface shall not exceed 25,000 Btu/hr per square foot of heating surface. The boiler heating surface shall consist of the surface of the heat transfer coils in contact on one side with water or wet steam, and on the other side with the combustion gas; this surface shall be measured on the side receiving heat.

3.8.4 Steam-pressure regulation. The control system shall maintain the steam pressure at the outlet connection within the following tolerance limits expected as a percent of the set point values of the operating control. The specified tolerances shall apply to any load which varies within 2 minutes from one-third of full load to full load and back to one-third of full load.

Fixed-rate, On-Off	± 15 percent
Three-position, High-Low-Off	± 15 percent
Modulating	± 10 percent

The class 1 and class 2 boilers shall produce steam in not more than 5 minutes from a cold start with a feedwater temperature of 60°F.

3.8.5 High-pressure hot water pressurization. The high-pressure hot water system shall be pressurized at least 20 psig above the maximum relative saturation temperature at the boiler discharge but not to exceed the design

Source: https://assist.dla.mil --Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

pressure specified in 1.2.1. The pressure drop through the heater coil shall not exceed 10 psig. The temperature at any point in the system shall be at least 20°F below saturation temperature.

3.8.6 Environmental conditions. Programming controls (see 3.9.1) shall be so designed that timing periods, exclusive of manufacturer's tolerances at rated operating temperatures, will not be affected more than 20 percent by variations of ambient temperature between 32°F and 120°F.

3.8.7 Noise. When specified (see 6.2.1), an assessment of the potential hazard of noise exposure at the site of the installation shall be performed by the contractor in accordance with ABMA, "The Measurement of Sound from Boiler Units, Bottom Supported, Shop or Field Erected." Equipment which produce sound pressure levels of 85 decibels on the A scale (dBA) on a standard sound level meter, dBA or greater shall be labeled with as 8- by 10-1/2-inch hazardous noise warning decal. Upon determination of the hazardous noise level of 85 dBA or greater, the contractor shall notify the Contracting Officer or his authorized representative. The Contracting Officer shall furnish two decals, NSN 0105-00-212-6020, for each boiler. The contractor shall affix one decal on each side of the boiler.

TABLE I. Requirements for combustion.

Fuel	Natural Gas 1,075 Btu/hr scf			Light Oils No. 2, JP-5 Diesel		
Percent of Rated load	40	70	100	40	70	100
Products of Combustion						
Carbon dioxide (minimum)	8.5	9.0 Note 1	10.0	10.5	11.5	12.5
Carbon monoxide (maximum)		Note 2		0.05	0.05	0.05
Excess air (maximum)	40	25	20	40	30	20
Equivalent Shell-Bacharach Smoke Scale (maximum)	0	0	0	2	2	2
Average Combustion Efficiency		80			81	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements for combustion. cont'd

Note 1	The values shown for carbon dioxide and oxygen are based on tests of a burner installed in a boiler. Values shown may be one percent higher for carbon dioxide and one percent lower for oxygen where air infiltration at the burner is minimized.
Note 2	Units rated below 20,000,000 Btu/hr input capacity shall meet UL 795 requirements for carbon monoxide. Units rated above 20,000,000 Btu/hr input capacity shall have no more than 0.05 percent carbon monoxide.

3.9 Control systems. Each boiler shall be equipped with the following control systems:

a. Solid-state programming control system: In conjunction with the electronic flame-failure control, safety interlocks and limit switches, the programming control system shall govern the prefiring, light-off, and shutdown cycle.
b. Combustion control system: This system shall automatically regulate the firing cycle in accordance with the load demand.

3.9.1 Programming control systems. The control sequences of the programming control system shall be as specified in 1.2.3 and as defined in NFPA 85A. The control sequences for the various sizes of burners along with their pertinent features and reference paragraphs are included. The basic programming and flame-failure controls shall be furnished as an integrated system by the boiler manufacturer and shall be designed to function in combination with other system components to insure the functional integrity of the total system. The program control system shall, as a minimum, denote by indicating lights the following occurrences: activation of power on or start of blower, pre-purge cycle, trial for pilot ignition, when main flame is detected, and when operating under automatic control. A post-purge period will not be mandatory on size 1 boilers, but shall be mandatory on size 2, 3, 4, and 5 boilers.

3.9.1.1 Control sequence IA. Control sequence IA shall be an automatic sequence (recycling) system having direct electric ignition for use with light oil size 1 boiler. Burners shall perform the following operations:

a. Manually operated switches shall apply power to the burner and energize the programming control.
b. Fan and timer shall start a forced-air purge of the combustion chamber of the furnace. The minimum purge shall be not less than 4 air changes and purge airflow rate shall reach not less than 60 percent of the flow rate at maximum output. For High-Low-Off combustion control, purge time may be 90 seconds when the airflow rate at low-fire start is not less than one-third maximum. All interlocks and electrical contacts, including a fan interlock switch, shall close or be closed before the end of the purge period and have indicated safe operating conditions (see 3.9.1.4).
c. After completion of the purge, the timer shall actuate the ignition, open the main fuel safety shutoff valve, or at the manufacturer's option, the valve opening may be delayed for a stipulated period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

d. When the main fuel valve opens, the main burner establishing period shall begin and activate a green indicating light (see 3.12.5). If the flame-sensing device does not sense the main flame within 10 seconds, a flame-failure safety shutdown (see 3.9.1.3) shall occur.

e. After the main flame is detected, the burner shall fire until the heat demand is satisfied. Upon satisfaction of heat demand, the burner shall shut off in accordance with the manufacturer's design procedure, and shall restart automatically when the demand is again established.

f. The burner shall establish safety shutdown automatically as a result of high coil thermostat temperature, low water flow, or high steam pressure, loss of control system actuating energy, flame failure (see 3.9.1.3), or other limit-switch, or low fuel pressure and air-flow safety-interlock opening as specified in 3.9.1.4. The safety shutdown shall accomplish the following in the order listed and will activate an alarm and a red indicating light, and deactivate the green light on the control panel.

(1) Shut off fuel supply to main burner.
(2) Interrupt spark ignition if in operation.
(3) Where used, open recirculating valve.
(4) Lock out programming control to require manual reset.

3.9.1.2 Control sequence IB. Control sequence IB shall be the automatic sequence (recycling) system with proved igniter for use on all gas, oil, or combination gas-oil burners. Unless otherwise specified in 3.11, burners shall perform the following operations:

a. Manually operated switches shall turn on the power to the burner and energize the programming control indicated by a blue indicating light (see 3.12.5). Switches shall start the circulation of the oil and recirculating system.

b. The prefiring cycle of the burner shall accomplish the following sequences:

(1) On boiler sizes 3, 4, and 5, prove the fuel safety shutoff valve is closed; for gas firing this shall be the downstream valve. On High-Low-Off combustion control systems the safety shutoff valve shall be located upstream in a line that is common to both the high and low fire fuel trains.
(2) Prove no flame present at the burner.
(3) Start fan.
(4) Satisfy airflow interlock.
** (5) Where an atomizing medium is used and if not already on, admit medium to main burner; on sizes 3, 4, and 5 activate a colored indicating light as indicated in 3.12.5.
** (6) Where atomizing medium is used, satisfy atomizing medium interlock.
** (7) Satisfy fuel pressure and limit switches and interlocks.
(8) Prove purge airflow by satisfying air pressure and "open damper" interlocks for all dampers in the flow path and satisfying all airflow interlocks. Purge air flow rate shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

reach not less than 50 percent of the flow rate at maximum output for boiler sizes 1 through 3, and not less than 20 percent for boiler size 4 and 5.

(9) The purge shall be sufficient for at least four air changes of the combustion chamber, heat exchangers, and flue passages. Airflow during the time to open and return the damper to light-off position may be included in computing the time required for the air changes.

(10) Set controls to light-off position (low-fire) for High-Low-Off and modulating combustion controlled burners.

(11) On units with High-Low-Off and modulating combustion controls, prove dampers, main flame fuel operational control valves in low-fire, and light-off position, respectively.

**NOTE: The sequential order of these items is optional; atomizing medium must be dry.

c. The light-off cycle of the burner, with the required interrupted igniter and all interlocks satisfied, (see 3.9.1.4), shall accomplish the following sequences in the order listed:

(1) Ignite igniter with ignition system as indicated in 3.11.
(2) Prove igniter flame within the igniter establishing period of 10 seconds.

(A) If proven, open the main fuel safety-shutoff valves to admit fuel to the burner. For an oil burner other than the return flow type, the burner shall simultaneously shut off oil recirculating flow.
(B) If not proven, an ignition-failure shutdown shall occur in accordance with 3.9.1.2(g).

(3) Prove main flame within the maximum time for main-flame establishing period of 10 second.

(A) If proven, shut off igniter, on igniters over 120,000 Btu/hr vent the gas igniter piping, release burner to operation cycle, and activate a green light in the control panel.
(B) If not proven, a flame-failure shutdown shall automatically occur in accordance with 3.9.1.2(g).

d. Operation cycle of the burner shall be accomplished by a combustion control system, as specified in 3.9.2.

e. Normal shutdown cycle shall be accomplished in the following order:

(1) Shut off fuel to the main burner.
(2) For oil:

(A) Where used, open recirculating valve.
(B) Where used, shut off atomizing medium, if desired.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

(3) For gas:

(A) Vent gas piping for boilers size 3 and larger between the main burner safety shut-off valves to atmosphere.

(4) After a post purge on size 2 boilers and larger if desired, shut down fan and deactivate green light in 3.12.5, otherwise fan shall remain running.

f. When the load demand is satisfied, automatic (recycling) burners shall accomplish a recycling shutdown (see 6.4.7), and the burner shall be allowed to recycle when the load demand is re-established within the preset operating range.

g. Nonrecycling safety-shutdowns (see 6.4.5) shall be accomplished in the following order and shall activate an audible alarm and a red indicating light (see 3.12.5).

(1) Shut off fuel supply to the igniter for ignition failure and main burner for main flame failure.
(2) Interrupt spark to the igniter, if in operation.
(3) Oil:

(A) Where used, open recirculating valve.
(B) Where used, shut off atomizing medium.

(4) Gas: Vent gas piping to igniter (as required) and main burner (see 3.10.2.1 and 3.11).
(5) After a post-purge on size 2 boilers and larger, shut down fan, if desired.
(6) Lockout programming control system (see 3.9.1.3) to require manual reset.

h. Any of the following conditions shall cause a nonrecycling safety-shutdown.

(1) High coil thermostat temperature.
(2) Low water flow.
(3) Power failure.
(4) Loss of combustion air supply.
(5) Loss of flame or failure to establish flame.
(6) Loss of control system actuating signal.
(7) For oil:

(A) Low oil pressure.
(B) Low temperature of heated oil.
(C) Loss of atomizing medium where used.

(8) For gas.

(A) High gas pressure.
(B) Low gas pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.9.1.3 Flame-failure control system. Each burner shall be equipped with a standard flame-failure control system (see 3.9). When specified (see 6.2.1), the flame-failure control system shall have a repetitive self-checking circuit and shall consist of a flame-sensing element properly connected to the programming control system for shutdown of the burner in the event of ignition failure or flame failure during the firing cycle. The burner shall not be allowed to recycle after ignition failure or main flame-failure without manual reset. The flame failure reaction time (including allowable factory tolerances) shall be not more than 4 seconds for all burners at rated control voltage in an ambient temperature of 70°F. Flame-sensing elements shall have solid-state amplifiers and shall not respond to ignition spark, hot refractory, or reflection of flame on atomizing media or oil spray. Flame-sensing elements for igniter flames shall detect only flames of sufficient intensity to safely light the main burner. Flame-sensing elements shall be so connected as to prevent startup if flame is present or if the contacts are stuck in the flame-sensing position. The flame-failure control circuit shall be so designed that the failure of any component prevents restart of the burner after shutdown, and shall incorporate automatic start-check at the beginning of each startup cycle. Flame-sensing elements shall consist of one type or a combination of the following types within the limitations indicated:

a. Ultraviolet cells may be used for gas or oil flames, but shall not be used for igniter proving unless: (1) the spark is shut off during the last part of the proving period, (2) the scanner is sighted so as to exclude spark detection, or (3) the scanner is equipped with a sight-tube to exclude spark detection.
b. Infrared (lead sulfide) cells may be used for gas or oil flame. Infrared cells shall not be used in locations where the cell could be actuated by residual radiation emanating from hot refractory after flame extinguishment.
c. Photolectric cell, rectification-type, for oil burners up to 3,500,000 Btu/hr.
d. Flame rods shall be used for gas burners up to 3,500,000 Btu/hr. Flame rods shall be of the rectification type, and shall be used only if the required length of the rod does not exceed 12 inches.

3.9.1.3.1 Repetitive self-checking circuit. The repetitive self-checking circuit shall automatically check all components of the flame-failure system including the flame detector. The component check of the detector is not required if the detector is fail-safe such as lead sulfide cells. During burner operation the self-checking circuit shall simulate failures at intervals not greater than 6 seconds. The occurrence of a flame-simulating failure anywhere in the flame-failure system or the cessation of the cyclic operation of the self-checking circuit shall cause a safety shutdown and lockout.

3.9.1.4 Safety interlocks and limit switches. Each boiler shall be equipped with the applicable safety interlocks and limit switches as required by the ASME Boiler and Pressure Vessel Code, ANSI CSD-1, and 3.9. The safety interlocks and limit switches shall sense the limit or off-limit conditions indicated, cause a nonrecycling safety shutdown or recycling shutdown of the burner in the specific cases annotated in the referenced documents, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

determine the position of the sensing element switch without disassembly of the device. All nonrecycling shutdowns shall require a manual reset of the interlock, limit switch, and primary flame safeguard control and shall be signaled by a red indicating light and an audible alarm. The safety interlocks and limit switches shall conform to UL 353. All safety interlocks and limit switches (including boiler and equipment limit switches) shall be wired in the hot side of the burner limit control circuit and not in any external circuits such as the pilot or fuel valve circuits.

3.9.1.4.1 Heating coil temperature control(s). Heating coil temperature control(s) shall cause a safety shutdown if the temperature of the heating coil exceeds the manufacturer's safe preset limit. The control(s) shall consist of a sensing device(s) required to prevent overheating of the coil from any possible cause, including an imbalance in circulating water input and steam output, or by partial or total interruption of water circulation.

3.9.1.4.2 Fuel safety controls. Fuel safety controls shall cause a nonrecycling safety shutdown if the oil pressure is below the safe preset limit, or if gas pressure is above or below the safe preset limits. In addition, depending on type and design of the burner, the boiler shall not operate if any of the following conditions will contribute to unsafe firing conditions when pressures are not within the limits determined by the manufacturer to be required for safe combustion:

a. Oil-atomizing steam pressure.
b. Oil-atomizing air pressure.
c. Primary air pressure for gas burners.

3.9.1.4.3 Combustion air safety control. The combustion air safety control shall function in a manner to cause a nonrecycling safety shutdown if insufficient air is available to effect the required purge in 3.9.1.1 or 3.9.1.2 on starting, or to support safe combustion during the firing cycle. The sensing device shall be an air flow, sail-switch type or a pressure-sensitive diaphragm type.

3.9.2 Combustion control systems. Combustion controls specified in 1.2.3 (except On-Off controls) shall automatically regulate the firing rate in accordance with the load demand and shall be capable of meeting the specified burner turndown (see 6.4.2) required in 3.10.1 or 3.10.2, as applicable. Control systems shall maintain air-fuel mixtures within the limits required for stable and efficient combustion throughout the specified firing range and, as applicable, during changes in firing rates. Control systems shall be of the fixed-rate, On-Off type; the three-position, High-Low-Off type; or the modulating type (see 1.2.3). Control systems for multi-unit boiler plants shall be as specified in the contract (see 6.2.1).

3.9.2.1 Fixed-rate, On-Off control. The fixed rate, On-Off control system shall operate to cycle the burner on and off in response to steam pressure for class 1 and 2 or water temperature for class 3 boilers. A fixed relationship shall exist between the volume of air and fuel admitted to the burner. The draft damper shall be manually adjustable to permit change of the fuel-air ratio when required by firing conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.9.2.2 Three-position, High-Low-Off control. The three-position, High-Low-Off control system shall automatically adjust the flow of fuel and air to the burner, as required, to provide either high or low fire in accordance with the load demand. The system shall function to cause a normal shutdown in response to a predetermined steam pressure (water temperature for class 3 boilers) below the steam safety valve or pressure relief valve setting when the load demand drops below the low fire output of the boiler and results in a buildup of pressure. The low firing rate shall be not more than 33.3 percent of the high firing rate.

3.9.2.3 Modulating control. The modulating control system shall be of the simple positioning type in which the volume of air and fuel delivered to the burner are varied simultaneously and gradually in accordance with the load demand while maintaining the proper air-fuel ratio. The control system shall include an adjustable direct positioning mechanism to maintain the proper ratio between the air supply opening and the area of the fuel valve opening. The minimum safe firing rate provided by the control system shall be not more than one-third of the maximum continuous firing rate at which the boiler is rated. The system shall function to cause a normal shutdown in response to a predetermined steam pressure (water temperature for class 3 boilers) below the steam safety valve or relief valve setting when the load demand drops below the minimum firing rate provided by the control system and results in a buildup of pressure.

3.10 Burners and fuel systems. The power burners shall be designed for the specified boiler type, class, capacity, and control systems (see 6.4.6).

3.10.1 Oil burners and fuel trains systems (type I boilers). Except as otherwise specified herein, oil burners and accessory fuel train components shall conform to the design requirements of UL 296 and ANSI CSD-1. Unless otherwise specified (see 6.2.1), burners shall be designed to fire grade No. 2 fuel oil conforming to the requirements of ASTM D396. Burners shall be mounted integrally with the boiler and shall be readily accessible for replacement, cleaning, and adjustment. Fuel oil trains shall be equipped, as required, to compensate for variations in oil supply pressure and to relieve excess pressure. The fuel trains shall include a manual shutoff valve, automatic fuel safety shutoff valves, oil strainer, pressure gage, fuel oil pump, and such flow rate control valves as may be required to meet the applicable requirements of 3.9.2. Wide-range mechanical pressure atomizing burners shall be of the return-flow type. A pressure gage shall be furnished with gage cock and with a scale 1.5 times the operating pressure. High-Low-Off and modulating oil burners shall have a burner turndown (minimum continuous safe firing rate), as specified (see 6.2.1), but shall be not more than 33.3 percent of the maximum continuous firing rate. Fuel trains shall withstand 1.5 times the operating pressure without rupture or leakage.

3.10.1.1 Fuel oil pump. The fuel oil pump shall be a positive or variable displacement, metering type. The pump shall have sufficient capacity to deliver not less than 120 percent of the volume of the specified grade of fuel oil required at maximum rated burner capacity. This capacity shall maintain 100 percent of the designed burner pressure measured at the burner tip when the oil supply system has a suction lift of not less than 10 feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.10.1.2 Strainers. Duplex, self-cleaning, or simplex strainers having bypass provisions as specified (see 6.2.1), shall be installed at the fuel pump suction connection and at such other locations as may be necessary to insure the delivery of clean oil to the burner.

3.10.1.3 Oil safety shutoff valves. The term "oil safety shutoff valve" as used herein shall be interpreted to mean an automatic valve which interrupts the flow of fuel to the burner on safety shutdown or on a normal shutdown after satisfaction of load demand as denoted in NFPA 85A. Dual safety shutoff valves shall be installed in series in the fuel supply piping. Group 5 boilers shall have a closed position interlock (over-travel) on the downstream safety shutoff valve. Oil safety shutoff valves shall be a positive-closing, spring-loaded, solenoid or motorized type. The time required for complete closing of the oil safety shutoff valves after the valves have been deenergized or activated for closing shall not exceed 1 second. Wide-range, mechanical, pressure-atomizing burners shall be equipped with a return-flow safety shutoff valve to prevent return oil leakage on safety shutdown. Steam atomizing and air atomizing burners shall be provided with a safety shutoff valve in the atomizing steam or air line. For automatic, nonrecycling units, one safety shutoff valve shall be of the manual reset type in the downstream position.

3.10.2 Gas burners and gas fuel systems (type II boilers). Except as may otherwise be specified herein, gas burners and accessory fuel system components shall conform to the design requirements of UL 795 and ANSI CSD-1. Unless otherwise specified (see 6.2.1), the gas heating values shown in 3.8, shall apply for the purpose of burner design and rating. The following conditions shall also be applicable unless the contract specifies alternate values (see 6.2.1):

a. Site elevation - 2000 feet or less.
b. Ambient operating temperature of 70°F ± 5°F.
c. 5 psig miminum gas supply.

The gas fuel supply system shall include a drip leg and a gas strainer. The gas main burner train shall be provided with a manually operated lubricated plug valve; pressure regulating valve; relief valve vented to atmosphere, automatic fuel safety shutdown valves; and flow-rate control valves as specified herein. Gas burners, when provided with the specified type of gas at the specified pressure, shall produce a stable flame which stays in one position relative to the burner face with no evident flashback into the burner and no evident blowoff from the burner face. There shall be no objectionable, pulsating noise or vibration of the flame to and from the burner face. The burner shall be so designed and located that all parts are protected against overheating. The valves shall be rated for a flow of the specified type of gas equal to at least 125 percent of the volume of gas required at maximum boiler output. The valves shall be suitable for the inlet gas pressure as specified (see 6.2.1) ahead of the gas regulating valve with a minimum gas supply pressure of 5 psig. A pressure gage with gage cock shall be furnished downstream of the safety shutoff valves with a scale 1.5 times the operating pressure. High-Low-Off and modulating gas burners shall have a burner turndown, as specified (see 6.2.1), but shall be not more than 33.3 percent of the maximum continuous firing rate.

Source: https://assist.dla.mil - Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.10.2.1 Gas safety shutoff valves. The term "gas safety shutoff valves" shall be interpreted in accordance with NFPA 85A. The gas safety shutoff valves shall have a UL label mark or FM approval symbol and conform to the following requirements.

a. Size 1, type II boilers shall be furnished with one main fuel solenoid operated safety shutoff valve, normally closed, electric reset and holding, valve to be spring closing and rated to close at a pressure higher than line pressure upstream from the pressure regulator. Valves shall be approved for gas service with a closing time not to exceed one second.

b. Sizes 2, 3, 4, and 5, type II boilers shall be furnished with two main fuel motorized safety shutoff valves, normally closed, electric reset and holding. On sizes 3, 4, and 5 boilers, the downstream valve shall have a closed position interlock. Valves shall be spring closing and rated to close at a pressure higher than line pressure upstream from the pressure regulator. Valve closing time shall not exceed one second on all burners. Valves shall have a leak test connection on the downstream side. Sizes 3, 4, and 5
type II boiler gas trains shall have a, general purpose, normally open, full ported electrically operated solenoid type vent valve, approved for gas service, that is connected in a manner to vent the piping between safety shutoff valve. When the safety shutoff valves are closed the piping shall be vented to atmosphere outside of building.

3.10.3 Combination gas and oil-fired units. A combination burner shall consist of an oil burner and a gas burner. Each shall conform to the requirements of this specification for separate oil and gas burners for the sizes of burners specified herein. The two burners shall be constructed as an integral combination unit suitable for firing either fuel separately or as separate quick-change burners. Suitable provisions shall be incorporated for withdrawing, shielding, or otherwise preventing the oil burner from coking while firing gas. Except as otherwise specified herein, the fuel systems and fuel safety interlocks shall consist of separate systems and components for each fuel. The air-supply and boiler-limit interlocks shall be common to all fuels when practicable. The flame-failure system components may either be separate for each fuel or common to all fuels in accordance with the particular design. The burner shall be so designed that all controls will be transferred from one fuel to the other by a single, manual selector switch furnished by the contractor. When the fuel is switched, the flame-failure, combustion, and fuel safety limit controls shall be automatically switched to the appropriate fuel. On fuel changeover, the programming control system shall recycle on gas and light oil.

3.11 Ignition systems. Boilers shall be equipped with interrupted ignition systems designed to supply sufficient ignition energy to provide positive ignition of all fuel discharged from the main burner under interlock, light-off conditions. Ignition timing for flame-establishing periods shall be as specified in 3.9.1.1 and 3.9.1.2. The ignition system shall be integrated with the programming control system specified herein. Gas igniter trains shall be equipped with a pressure regulating valve, relief valve, bleed

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

valves, approved solenoid-operated safety shutoff valves, and lubricated manual shutoff valves in accordance with ANSI CSD-1 for boilers up to 12,500,000 Btu/hr and NFPA 85A for boilers greater than 12,500,001 Btu/hr input capacity. Oil igniter trains for size 2 burners in (c) and (d) below shall include a strainer (either separate or integral with the nozzle), filter, and two solenoid-operated safety-shutoff valves. If the oil igniter system includes a pump (separate from the main fuel oil pump), which shuts down in response to the safety control circuit, only one igniter safety shutoff valve will be required. Oil igniters shall be of the straight mechanical or low-pressure, air atomizing type designed for firing No. 2 fuel oil. Ignition transformers shall be rated on the secondary side at not less than 6000 volts (V) for gas-electric igniters and 10,000V for oil-electric igniters with dual electrodes. A normally open valve (UL labeled for gas service), equipped for automatic closing and spring opening shall vent the igniter gas piping to the atmosphere (outside of the installation room) between the two safety-shutoff valves if the igniter is rated in excess of 120,000 Btu/hr. Burners shall be equipped with ignition systems conforming to the following requirements:

a. Except as otherwise specified herein, all igniters shall be the interrupted type with gas-electric ignition.
b. On size 1 burners firing No. 2 fuel oil, direct electric spark ignition may be provided.
c. When specified (see 6.2.1), direct spark ignition shall be furnished on oil-fired burners in lieu of gas-electric igniters of the interrupted type on burners having High-Low-Off combustion control systems with 20 gallons per hour limit on light-off where gas is not available.
d. When specified (see 6.2.1), light oil spark ignition igniters of the interrupted type shall be furnished to light the main flame on size 3, 4, and 5 oil-fired boilers. Igniters and piping shall be in accordance with NFPA 85A where gas is not available.

3.12 Components and accessories.

3.12.1 Pumps. Pumps shall be furnished, as required by the manufacturer's standard design, to provide a flow rate through the heating coil sufficiently in excess of the evaporation rate. This is done to insure that the steam temperature does not exceed the saturation temperature for any conditions of steam output and pressures for which the boiler is designed specifically for modulating burner controls. The pump(s) shall be electric motor driven. The feedwater inlet piping shall include a suction strainer, manual shutoff valve, and check valve in the pump discharge line.

3.12.2 Draft fan. The draft fan shall be direct-connected to or belt-driven by an electric motor. The fan and motor shall be sized for an output volume and static-pressure rating sufficient for complete combustion of the maximum fuel to be burned plus 10 percent excess volume against a 20 percent static pressure overload. Provisions shall be made to vary the fan discharge rate for burners with modulating controls. Mechanisms shall operate smoothly without backlash. Fan noise shall not exceed 90 dBA on a standard sound level meter at 3-foot stations at a height of 4.5 feet.

Source: https://assist.dla.mil - Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

3.12.3 Stack. When specified (see 6.2.1), stack sections, stack supports, umbrella collar and cap, and flue transition piece (if required) shall be provided. Unless otherwise specified (see 6.2.1), the stack shall be 20 feet in height when assembled on the boiler and measured from the ground line. Stack sections shall be sheet steel having a thickness of not less than 0.0972 inch or UL labled multi-wall, pre-fabricated construction with corrosion resistant steel of not less than ASTM A176-UNS 541000. When required, flue transition pieces shall be furnished to connect from the flue outlet to the stack. The transition piece shall be capable of carrying the full weight of the stack. When specified (see 6.2.1), a slip-type expansion joint shall be supplied for installation at the flue outlet to allow for expansion of the stack.

3.12.4 Condensate return system. When specified (see 6.2.1), a condensate return system shall be furnished with each steam unit. The system shall conform to type I of MIL-P-17749 and shall be sized to deliver not less than twice the total feedwater required at maximum rated output of the unit. The capacity of the circular condensate receiver shall be not less than one-fifth the volume of water converted to steam per hour.

3.12.5 Control panel. The controls, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and circuit elements of control systems shall be mounted on a single control panel or cabinet insofar as practicable and shall be designed for a separate mounting not on the burner. Switches shall include burner start, manual/automatic combustion control, fuel oil heater, and fuel selector switches, as applicable. Colored indicating lights shall be provided and shall be activated when the interlock is satisfied for the control sequences herein and shall be permanently identified. Unless otherwise specified (see 6.2.1), the number and colors of lights shall indicate operations as follows:

a.	High-heat exchanger coil temperature	Red
b.	Low water flow	Red
c.	Safety lockout	Red
d.	Atomization, medium, when used	Yellow
e.	Auto control	Green
f.	Power on	Blue

The control panel or cabinet or the individual elements shall be provided with NEMA ICS 6 type 12 enclosures or when specified (see 6.2.1), NEMA ICS 6 type 3R enclosures. The control panel shall be mounted on a solid base. All terminals requiring connection upon installation of the contract panel shall be permanently identified in such a manner that correct connections can be easily made by reference to the instructions and wiring diagrams provided with the equipment. Terminal connections for control functions shall be segregated from power circuit terminals. The control panel shall be located in a safe, nonhazardous location with the burner visible from the operator's stations to protect the operator from flareback. Burner and equipment controls, including the flame and igniter failure safeguard controls, shall be mounted in the control panel attached to the side of the boiler (right or left side when facing the equipment), for wall mounting, or for remote mounting in a free-standing control panel as specified (see 6.2.1). When a free-standing control panel is to be furnished, construction details covering cabinet

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

location, mounted controls, wiring for remote starting, and instruments shall be as specified (see 6.2.1 and 6.2.2). The flame safeguard and burner controls shall not be mounted on the burner. When specified (see 6.2.1), a main circuit breaker shall be furnished and mounted either adjacent to the control panel or packaged separately for remote mounting at the site.

3.12.6 Annunciator. When specified (see 6.2.1), an annunciator shall be provided for boilers, sizes 3 through 5. Unless otherwise specified (see 6.2.1), the annunciator shall indicate all nonrecycling burner shutdowns specified in 3.9.1.2, including operational control alarms, and provide at least two spare stations or points in addition to those specified herein. The annunciator shall not require special or additional limit devices, change sensitivity of limit switches, or annunciate without the limit device being actuated. The annunciator shall provide means to indicate the first-limit or off-limit condition causing burner shutdown and shall sound an alarm. The annunciator shall also have a manual reset button and capability to store information on additional causes of burner shutdown. When specified (see 6.2.1), subsequent causes of burner shutdown will be indicated after the operator acknowledges the first burner shutdown. A dusttight enclosure shall be provided in accordance with NEMA ICS 6 type 12 or when specified (see 6.2.1) NEMA ICS 6 type 3R. The annunciator shall include, but not be limited to, additional features and information required in Section 5 of ISA S18.1, as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), the annunciator may either be a separate item or an integral part of the control panel or burner control system.

3.12.6 Smoke density indicator and transmitter. When specified (see 6.2.1), an electronic opacity monitoring device with strip chart recorder and transmitter for stack mounting shall be provided to detect, indicate, sound an alarm, and record visible emissions from the stack. The system shall conform to the requirements of CFR Title 40, Part 60.344, Method 9, "Visible Emissions." A scale indicator shall be provided to display the output of the smoke density transmitter calibrated in percent opacity. The instrument need not have a clean air, self-cleaning feature. The instrument shall operate in normal 120V (minimum 102V, maximum 132V), 60 cycles per second, and have internal electronic voltage regulation with power consumption of 20 amperes maximum. The smoke density indicator and transmitter locations shall be as specified (see 6.2.1).

3.13 Electrical requirements.

3.13.1 Motors. Motors for the driven auxiliaries shall be constructed and rated in accordance with the applicable provisions of NEMA MG 1. Motors shall be general-purpose, continuous duty, ball-bearing, induction motors. The motors shall be designed for operation on a power supply having the specified characteristics (see 6.2.1). Unless otherwise specified (see 6.2.1), motors shall be furnished with drip-proof enclosures. When specified (see 6.2.1), motors shall be totally enclosed fan cooled.

3.13.2 Motor starters. Unless otherwise specified (see 6.2.1), motors less than 15 hp shall be provided with magnetic across-the-line starters with overload protection and manual-off-automatic control stations. When specified (see 6.2.1), motors over 15 hp shall have reduced voltage starting with

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

overload protection and manual-off-automatic control station. Magnetic starters shall conform to the applicable requirements of NEMA ICS 1, ICS 2, and ICS 6, and shall have NEMA ICS 6 type 12 enclosures or when specified (see 6.2.1), NEMA ICS 6 type 3R enclosures.

3.13.3 Wiring and safety devices. All electrical wiring and circuit safety devices shall be suitable for operation at a minimum of 40 degrees Centigrade (°C) and shall conform to UL 726, UL 795, and NFPA No. 70 as applicable. All control circuits shall be two-wire, one-side grounded, not to exceed nominal 120V, and protected with suitable fuses or circuit breakers. All safety interlocks and limit switches shall be wired in the hot side of the burner limit control circuit and not in any external circuits such as pilot or fuel valve circuits. When a free-standing control panel is required (see 3.12.5), automatic sequence burners shall be wired for remote starting from the panel.

3.14 Lubrication. Unless otherwise specified (see 6.2.1), the means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated prior to delivery. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psig or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.15 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be treated for moisture and fungus resistance in accordance with MIL-T-13867. Electrical components with a temperature rise of 40°C or less shall be coated in accordance with type II of MIL-T-13867 with fungus-resistant varnish conforming to MIL-V-173. Components with a temperature rise exceeding 40°C shall be coated in accordance with type III or IV of MIL-T-13867 with varnish conforming to MIL-I-24092 with grade and class of varnish dependent on the component, class of insulation above 40°C, and the recommendations of the manufacturer of the equipment.

3.16 Electromagnetic interference suppression. When specified (see 6.2.1), the unit shall conform to the electromagnetic interference suppression requirements and test limits for class 3, group III, equipment as specified in MIL-STD-461.

3.17 Lifting attachments. Suitable lifting attachments and tiedown devices shall be furnished in accordance with the manufacturer's standard prctice except as specified herein. The attachments and tiedown devices shall have eyes not less than 3 inches in inside diameter and plates to withstand any handling conditions encountered. The tiedown devices shall be located in accordance with the manufacturer's standard practice. Adequate tiedown devices may also be used for lifting purposes. Each attachment or tiedown device shall be of sufficient capacity to carry the total weight.

3.18 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.19 Technical manuals. The contractor shall provide technical manual(s) conforming to the Data Item Description(s) (DID) listed on the Contract Data Requirements List (CDRL) (DD Form 1423) (see 6.2.2).

3.20 Instruction plates. Unless otherwise specified (see 6.2.1), the unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.21 Identification marking. Identification shall be permanently and legibly marked directly on the burner or on a corrosion-resisting metal plate securely attached to the boiler at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer.

3.21.1 Military marking. When specified (see 6.2.1), an identification plate will be furnished by the contracting officer for each boiler. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose and securely affix a plate to each boiler in a conspicuous place with nonferrous screws or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.22 Workmanship. Workmanship shall meet the quality standards prevailing among recognized manufacturers of steam generating equipment and as established by the ASME Boiler and Pressure Vessel Code, UL 296, UL 726, and UL 795.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. On-site testing (see 4.2.3).

4.2.1 First article inspection. The first article inspection shall be performed on one unit when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 6.2.1). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3; the tests of 4.4.1, 4.4.2.1, and 4.4.4; and the packaging inspection of 4.5.

4.2.3 On-site testing. Testing shall be performed on the boiler(s) at the site after installation and shall be witnessed by a certified boiler inspector acceptable to the Contracting Officer. This testing shall be in addition to the quality conformance inspection performed at the factory and shall be supplemental to first article inspection, if such inspection is specified (see 6.2.1). On-site operational testing shall consist of tests of 4.4.2 and all tests deemed necessary by the acquisition activity to verify compliance with the requirements of this specification. On-site testing shall be performed either by the boiler manufacturer or by the installing contractor, as specified (see 6.2.1). The manufacturer shall have the privilege of representation at tests performed by others. When the manufacturer is responsible for on-site tests, the detailed requirements and schedule for the test program shall be as specified in the contract (see 6.2.1). In all cases, deficiencies revealed by on-site testing shall be corrected at the boiler manufacturer's expense and any required retesting shall also be at the manufacturer's expense.

4.3 Examination. Each boiler shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

4.3.1 Design verification. When specified (see 6.2.1 and 6.2.2), the contractor shall furnish drawings and computations necessary to verify compliance with the heat transfer rate of 3.8.3.

4.3.2 Standards compliance. When specified (see 6.2.1), the contractor shall certify (see 6.2.2) to the Contracting Officer or his authorized representative compliance with standards cited in 3.10.1, 3.10.2, 3.10.2.1, 3.10.3, and 3.11 as applicable. The Government reserves the right to examine and test all units to determine the validity of the certification.

4.4 Tests. Tests shall consist of the following:

a. Hydrostatic tests (see 4.4.1).
b. On-site operational tests (see 4.4.2).
c. Fuel-to-steam and fuel-to-hot water efficiency tests (see 4.4.3).
d. Auxiliary equipment tests (see 4.4.4).
e. Electromagnetic interference suppression tests, when required (see 4.4.5).

The hydrostatic (see 4.4.1) and on-site operational tests (see 4.4.2) shall be performed on each boiler supplied under a contract, including the first article, if furnished. Capacity and efficiency tests and auxiliary equipment tests shall be performed only on the first article. The electromagnetic suppression tests shall be performed on one unit when suppression control is specified (see 6.2.1).

4.4.1 Hydrostatic tests.

4.4.1.1 Watersides. Watersides shall be hydrostatically tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code. The appropriate symbol stamps specified in 3.7 shall be accepted as evidence of compliance.

4.4.1.2 Fuel systems. The preassembled fuel system furnished integrally with the boiler shall be checked for leakage during the fire testing of the boiler. The soap bubble or the halogen sniff test method shall be used to verify integrity of the gas system. The fuel oil system shall be tested in accordance with 3.10.1.

4.4.2 On-site operational tests. Each boiler shall be fire-tested to demonstrate control and operational conformance and combustion efficiency to the requirements of 3.8.2.1, 3.8.4, and 3.8.5 and to the requirements of this specification under varying load conditions ranging from the specified capacity to the minimum burner turndown ratio without on-off cycling. All burners shall be tested for the ability to start a cold boiler in accordance with the manufacturer's instructions. Boilers that exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected. Operational tests shall be as specified in 4.4.2.1 through 4.4.2.12.

4.4.2.1 Capacity, quality of steam, and efficiency fire-tests. Input capacity specified in 3.8.1 shall be measured and recorded. The moisture content of the steam shall be as specified in 3.7.3. Efficiency at the

Source: https://assist.dla.mil -- Downloaded:.2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

capacity of all boilers shall be determined by an analysis of the flue gases in accordance with ASME PTC 4.1. Performance shall be as specified in 3.8.1, 3.8.2.1, and table I. Test runs shall be made at maximum capacity and for units with High-Low-Off at the lower limit of the burner turndown. Six samples of the combustion gases for each load setting shall be drawn evenly distributed across the face of the flue exit. Since the relation between carbon dioxide and oxygen is both interdependent and specific to the fuel type, measurements of both gasses are required to identify infiltration problems. A variation in either flue gas value of over 2 percent may be an indication of air infiltration. Testing should be terminated until the cause of the variance has been identified and the problem corrected.

4.4.2.2 Combustion control tests. The boiler shall be tested under varying load conditions to verify compliance with the requirements for combustion controls as specified in 3.9.2. The accuracy range and smoothness of operation of the combustion controls shall be demonstrated by varying the load demand through the entire firing range required by the specified burner turndown, and in the case of On-Off controls, by further varying the firing rate to require On-Off cycling. The control accuracy shall be as specified in 3.8.4. Boilers which exhibit excessive roughness of operation of the combustion controls or difficult adjustment shall be rejected.

4.4.2.3 Spark rejection test. With the fuel supply to both the igniter and main flame closed, the burner shall be operated in the ignition cycle. When the spark ignition comes on, the flame detector shall not be activated. If the flame detector can be activated, the burner shall be rejected. This requirement does not apply for direct spark ignited burners.

4.4.2.4 Burner turndown. The specified burner turndown for High-Low-Off and modulating controls shall be verified by firing at the minimum firing rate required by the specified burner turndown (see 3.10.1 and 3.10.2).

4.4.2.5 Smoke test. The smoke density reading shall be determined in accordance with ASTM D2156 and shall not exceed the smoke scale numbers at the percentages of rated load in table I.

4.4.2.6 Sequencing. The boiler shall start, operate, and stop in strict accordance with the specified operating sequence of 3.9.1.

4.4.2.7 Flame failure. The operation of the flame failure control shall be verified by simulated flame and ignition failures. The igniter establishing period, the main-burner establishing period, flame-failure reaction time, and the safety-shutoff valve closing times shall be verified by stop watch (see 3.9.1.1, 3.9.1.2, and 3.9.1.3).

4.4.2.8 Immunity to hot refractory. The boiler shall be operated at high fire until the combustion chamber refractory reaches maximum temperature. The main fuel valve shall then be closed manually. The combustion safeguard shall drop out, causing the safety shutoff valves to close within the specified flame-failure reaction time.

4.4.2.9 Minimum igniter test. The fuel supply to the igniter flame shall be gradually reduced to the point where the combustion safeguard begins to drop out (sense "no flame"). At this point of reduced igniter fuel supply,

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

the igniter flame shall be capable of safely igniting the main burner. If the main fuel valve can be opened on an igniter flame of insufficient intensity to safely light the main flame, the burner shall be rejected.

4.4.2.10 Boiler limit and cutout switches and fuel safety interlocks. Safety shutdowns shall be caused by simulating interlock actuating conditions for each of the boiler limit and cutout switches and fuel safety interlocks. Safety shutdowns shall occur in the specified manner.

4.4.2.11 Safety valves. The high-pressure limit switch shall be locked out or otherwise made inoperative and the boiler safety valves shall be lifted by steam. The relieving capacity, popping pressure, blowdown, and reseating pressure shall be determined by observation and measurement to be in accordance with the ASME Boiler and Pressure Vessel Code. This test shall be made in the presence of the Contracting Officer or his authorized representative. The ASME standard symbol will be accepted only as indicating compliance with the design and material requirements of the code.

4.4.2.12 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

4.4.3 Fuel-to-steam and fuel-to-hot water efficiency tests. The fuel-to-steam and fuel-to-hot water efficiency, and steam quality at the specified capacity of the generator shall be determined in accordance with ASME PTC 4.1. The efficiency shall be determined by the input/output method. Performance shall be as specified in 3.8.2.2 and 3.8.2.3. Test runs shall be made at the maximum rated capacity.

4.4.4 Auxiliary equipment tests. Draft fans, fuel oil heaters, fuel pumps, dampers, actuators, valves, and electric motors shall be operationally tested to determine compliance with the referenced specifications or standards. Standard approval symbols or certificates of compliance (see 6.2.2) from certifying organizations may be accepted at the discretion of the Contracting Officer or his authorized representative. The operation of fans, fuel oil heaters, fuel pumps, and electric motors shall be closely observed during the on-site operational tests (see 4.4.2) for possible defects or nonconformance. The action of dampers, actuators, and valves shall be smooth and without backlash.

4.4.5 Electromagnetic interference suppression tests. To determine conformance to 3.16, the first article shall be subjected to the tests in accordance with MIL-STD-462. The manufacturer upon approval of the Contracting Officer may furnish a certification (see 6.2.2) in lieu of the test that the boiler meets the requirements of 3.16, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180. The inspection shall consist of the quality conformance inspection. When specified (see 6.2.1), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2.1).

Source: https://assist.dla.mil Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with MIL-B-3180 with the level of preservation and level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The boilers covered by this specification are intended for use in the generation of steam and hot water for processing operations, steam heating, or powerplant standby service where economics, space limitations, and operating conditions justify their use. The steam boilers have a high specific steam output per cubic foot of volume and are, therefore, used to advantage where space is limited and for mobile or portable installation on trucks, trailers, etc. The fast steaming rate (see 3.8.4) increases the adaptability of the boilers to powerplant application for emergency standby service and the handling of off-peak loads.

6.1.1 Selection factors. The following factors should be considered in the selection of a boiler:

a. With the present solid-state, programming control reliability, automatic (nonrecycling) boilers should only be specified when automatic boilers are not desired.
b. Repetitive, self-checking circuits should not be included as standard because of the frequency of start and stop operation (once in approximately every 5 minutes) and only should be included where boiler in continuous operation.
c. The boiler type (oil, gas, or combination) should be determined in accordance with availability or economics of the fuel supply and any Department of Defense requirements for capability of firing multiple fuels.
d. Control sequence 1A should be specified for size I type I light oil.
e. High-Low-Off and modulating combustion controls should be specified for size 2 and larger.
f. The steam or high-pressure hot water boilers should be sized to permit efficient boiler operations under low load conditions during summer.
g. The temperature at any point in the circulating system for high-pressure hot water boilers should be at least 20° F below the saturation temperature for the pressure at that point.
h. Gas/electric igniters are required for UL, FM, and Industrial Risk Insurers. Direct spark igniters or oil igniters should be specified only when gas is not available.
i. A smoke density indicator should be specified on all oil-fired and combination boilers over 5,000,000 Btu/hr input capacity.

6.1.2 Other fuels. The performance requirements specified herein may be based on other fuels providing the quantity on-site is sufficient for tests specified in section 4 and a guaranteed ultimate analysis, higher heat content, and oil viscosity are made available at time of testing. Corrections between the values of the analyses in section 3 of this specification and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

guaranteed analyses and heating value should be made in accordance with section 7 of ASME PTC-4.1. Referenced below are some of the most commonly used fuels with their average viscosity and higher heat content:

Fuel	Average viscosity	Average higher heat content
Light Oil		
Grade 2 and 4 ASTM D396 JP-5 MIL-T-5624 Diesel VV-F-800	125 Saybolt Universal Seconds(SUS)	140,000 Btu/gal
	Specific gravity	
Natural gas	0.65	1075 Btu/scf
Manufactured gas	0.38	535 Btu/scf
Mixed gas	0.50	800 Btu/scf
Liquified petroleum gas	1.55	2500 Btu/scf

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, date, and number of this specification.
b. Type, class, size, and actual capacity of boiler required in Btu/hr, design pressure for class 2 and 3; grade of fuel oil or type of gas to be burned, type of programming control system, and type of combustion control system (see 1.2.1, 1.2.2, 1.2.3 and 3.8.1).
c. If specifications and standards are different (see 2.1.1).
d. When first article is required (see 3.2, 4.2.1 and 6.3).
e. The following design conditions are required (see 3.4):
   (1) Boiler design pressure (psig) for class 2 and class 3 boiler (see 1.2.1).
   (2) Boiler operating pressure (psig) (see 3.4.).
   (3) Steam temperature (°F) (see 3.4).
   (4) Feedwater temperature (°F) (see 3.4).
   (5) Temperature differential (°F) for hot water boiler between boiler discharge and system return (°F) (see 3.4).
   (6) Maximum and minimum anticipated ambient air temperature (°F) (see 3.10.2).
   (7) Boiler site elevation (feet) (see 3.10.2).
   (8) Maximum continuous output in Btu/hr and if on a different bases (see 3.8.1).
f. When moisture content of the steam is other than as specified (see 3.7.3).
g. When boiler water concentrations are other than specified (see 3.7.3).
h. When a feedwater pump is not required with drum-type separators (see 3.7.3.2).
i. Special sizes and types of steam connections, if required (see 3.7.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

j. When performance criteria shall be based on fuel characteristics other than as specified. Provide the required fuel characteristics (ultimate analysis) and higher heating value of intended oil or gas fuel if different (see 3.8, 3.10.1, and 3.10.2).
k. Available gas supply pressure at site upstream of the service pressure regulator (see 3.8).
l. When an assessment of the potential hazard of noise exposure is required (see 3.8.7).
m. When a repetitive self-checking circuit is not required with the flame-failure control system (see 3.9.1.3).
n. Requirements for control systems for multi-unit boiler plants (see 3.9.2).
o. Required burner turndown expressed as percentage of maximum continuous firing rate (see 3.10.1 and 3.10.2).
p. Type of stainer(s) required (see 3.10.1.2).
q. When valves shall be suitable for pressure greater than 5 psig minimum (see 3.10.2).
r. When direct spark ignition shall be furnished on High-Low-Off combustion control systems with 20 gal/hr limit on light-off (see 3.11(c)).
s. When light oil spark ignition igniters shall be furnished to light the main flame on size 3 boiler and larger (see 3.11d).
t. When a stack is required and the stack height is other than as specified; when a slip-type expansion joint is required (see 3.12.3).
u. When a condensate return system is required (see 3.12.4).
v. When number and colors of lights are different on control panel; when NEMA ICS 6, type 1A cabinet is required; locations of control panel; construction details, when free-standing control panel is required; when main a circuit breaker is required (see 3.12.5).
w. When an annunciator is required for size 3 through size 5 boilers and if different from as specified; when NEMA ICS 6 type 3R enclosure is acceptable, when annuciation is required for subsequent causes of burner shutdown, additional features and information required in accordance with section 5 of ISA S18.1; whether annunciator is to be a separate item or an integral part ot the control panel (see 3.12.6).
x. When a smoke density indicator and transmitter is required and location of each (see 3.12.7 and 6.1.1b).
y. Characteristics of the electrical power supply available at the site and if motor enclosures are different; when motors are to be totally enclosed fan cooled (see 3.13.1).
z. If motor starters are different and when NEMA type 3R enclosures are required; when motors over 15 hp shall have reduced voltage starting with overload protection and manual-off-automatic control station (see 3.13.2).
aa. If lubrication is different (see 3.14).
bb. When treatment for fungus resistance is required (see 3.15).
cc. When electromagnetic interference suppression is required (see 3.16 and 4.4).
dd. Required color of finish coat (see 3.18).
ee. If instruction plates are different (see 3.20).
ff. When military marking is required (see 3.21.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

gg. When an on-site inspection is required, whether on-site inspection will be performed by the manufacturer or by the installing contractor; outline of tests if boiler manufacturer will perform the tests (see 4.2.3).
hh. When drawings and computations are required to verify compliance with heat transfer rate (see 4.3.1).
ii. When the contractor is required to certify compliance with standards cited (see 4.3.2).
jj. When a preproduction pack is required and the time frame required for submission (see 4.2.1 and 4.5).
kk. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, CDRL, the data requirements identified below shall be developed as specified by an approved DD Form 1664, DID, and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No	Data requirements title	Applicable DID No	Option
3.3, 4.3.2, 4.4.4, 4.4.5	Certificate of compliance	DI-E-2121	None
4.3.1	Design Data and calculations	DI-R-24040A	None
3.12.5	Drawings, Engineering and Associated Lists	DI-E-7031	None
3.19	Manual with supplementary data, technical commercial equipment	DI-M-2050	None

(DIDs related to this specification and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete boiler. The Contracting Officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Definitions. The following definitions are applicable for use by the contractor and acquisition agency in the interpretation and implementation of the requirements of this specification. Other terms used in this specification should be interpreted in accordance with the definitions of UL 296, UL 726, UL 795, and the ABMA Lexicon.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

6.4.1 Burner control system. A burner control system is the system by which a furnace is purged and the burner is automatically started, ignited, monitored, and stopped. Automatic, recycling burners restart without manual actuation when a preset pressure or temperature is reached. Automatic, nonrecycling burners shall require manual actuation of a starting switch and fuel shutoff valve to reestablish the firing cycle.

6.4.2 Burner turndown. The burner turndown shall be the minimum continuous safe firing rate which can be effected automatically by the combustion control system during the normal firing cycle without interruption or change of burner components. The burner turndown shall be expressed as a percentage of the maximum continuous firing rate.

6.4.3 Flame inpingement. Flame impingement means physical contact of the burner flame on the surface (fireside) carrying the products of combustion of the furnace or any tube and tube sheet of the boiler so that the flame path is diverted at these surfaces.

6.4.4 Input capacity. For this specification, the input capacity of a boiler is the product of the volume of fuel consumed per hour, and the chemical heat content of the fuel per volume (higher-heat value of the fuel per volume). The site elevation and temperature conditions shall be as specified in 3.10.2.

6.4.5 Nonrecycling safety-shutdown. Stopping burner operation by shutting off all fuel and ignition energy to the furnace by means of a safety interlock or safety control and requiring a manual reset.

6.4.6 Power burner. A burner in which oil or gas is supplied individually along with air at pressures; for oil, line pressure; for gas, exceeding line pressure; and for air, exceeding atmospheric pressure, this added pressure being applied at the burner.

6.4.7 Recycling shutdown. Stopping of a burner operation by a normal shutdown when certain interlocks and limit switches have sensed a limit or off-limit condition and then allows the burner to automatically recycle when the condition corrects itself.

6.4.8 Trial-for-ignition period. (Main Burner Establishing Period). That interval of time during light-off cycle in which a safety control circuit permits the main burner fuel safety shutoff valves to be opened before the flame detection system is required to supervise the main burner flame.

6.5 Cross-reference. The following cross-reference of classification applies.

MIL-B-17228D	MIL-B-17228E
Type I	Type I
Type II	Type II
Type III	Type III
Class 1	Class 1
Class 2	Class 2
Not designated	Class 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

Group 1 and Group 2	Size 1
Group 2 and Group 3	Size 2
Group 3	Size 3
Group 3	Size 4
Not designated	Size 5

Programming Control Systems

Automatic recycling:	
With direct electric ignition	Control sequence IA
With pilot ignition	Control sequence IB
Automatic nonrecycling:	
With direct electric ignition	Deleted
With pilot ignition	Control sequence II

Combustion Control Systems

On-off	On-off
High-Low-Off	High-Low-Off
Modulating	Modulating

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review Activity

Air Force - 99

User Activities:

Army - CE
Navy - MC

Preparing Activity:

Navy - YD

(Project 4410-0045)

Source: https://assist.dla.mil - Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-17228E	Boilers, Steam and Hot Water; High- and Low-Pressure,

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:52Z
Check the source to verify that this is the current version before use.